Bank early warning governance memo

FINTECH 590 Phase 3 | Team policy design | Model ews-xgb-0.2

Recommendation

We recommend a limited quarterly credit, earnings and capital triage pilot only after independent human validation. The team retains the current target and review capacity for Phase 3. Version 0.2 corrects the forward-label defect, but its 7.53 percent recall at the fixed cutoff and its failure to flag the three 2023 cases do not support a general bank-failure predictor. Require a separate liquidity review and prohibit autonomous decisions. [S4, S8]

Intended use and responsibility

A flag requests investigation of a Call Report filing bank; it does not determine action. Already-breached banks need a separate list. Proposed roles are a pipeline owner who produces versioned results, a reviewer independent of model construction who checks outputs and proposes exceptions, and a committee chair who approves use and overrides. These controls are proposed, not roles already staffed.

Failure mode 1 Liquidity events are outside the target

Evidence: at 2022 Q4, v0.2 leaves SVB, Signature and First Republic unflagged at ranks 2,208, 3,783 and 2,832. The target covers capital, noncurrent loans and earnings, rather than a liquidity-run event. The Federal Reserve describes concentrated funding and interest-rate risks in the 2023 failures. [S1, S7, S8]

Control: review available funding concentration, securities losses and dated liquidity evidence separately. Add banks to review regardless of model score when that evidence warrants it. A low score must never clear a liquidity concern.

Failure mode 2 Incomplete follow up distorts labels

Evidence and correction: v0.1 could mark partial non-breach follow-up as negative. In v0.2, negatives require all four actual future calendar quarters; 16,450 labels become unresolved and no original positives change. First Republic at 2022 Q4 changes from 0 to unresolved. SVB and Signature were already unresolved in v0.1. [S2, S8]

Control: retain regression tests for calendar gaps, panel exits and recent immature quarters. Distinguish missing follow-up from a healthy outcome. The corrected labels match the ingested panel in automated review, but that check does not independently validate every raw filing or identify why a bank stopped reporting.

Further failure modes and controls

Concentration and interpretation

Failure mode 3 One bank dominates deposit coverage

Evidence: BofA NA is flagged at rank 54 with score 0.9141. Its leverage of 7.1144 percent is the largest positive SHAP driver. Flagged banks hold 10.36 percent of system deposits; excluding BofA, only 0.12 percent. Being close to the 7 percent target line is a review concern, not proof that changing that line would remove its flag. [S1, S8]

Control: show both total coverage and concentration, keep overflow flags visible, and review BofA with source evidence. Test alternative target definitions by rerunning the model before claiming sensitivity. Do not lower a cutoff simply to remove an inconvenient bank.

Failure mode 4 Scores are mistaken for probabilities

The positive class is reweighted in training and calibration has not been established. Label outputs risk score; a value of 0.90 must not be described as a 90 percent event probability or used directly in expected-loss calculations. Require a later matured calibration study before relaxing this restriction. [S3, S4]

Failure mode 5 Evaluation is reused for selection

The 2023 to 2024 evaluation compares model variants and supplies the cutoff. We retain deduplicated 12 features for broader driver coverage while disclosing that naive 12 has better average precision, 0.2208 versus 0.2077. The comparison is development evidence, without uncertainty bounds or an untouched final test. Reserve a later matured holdout or a documented rolling-origin design before operational approval. [S3, S8]

Failure mode 6 Reporting proxies alter comparisons

The pipeline NII ratio uses total assets, ROA uses quarter-end assets, and CRE to capital can use a Tier 1 fallback. Keep those definitions visible and inspect availability and performance by reporting form and size. A size-related score pattern alone does not establish either economic risk or model bias. Missing trigger inputs also need an explicit eligibility check. [S2, S7]

Failure mode 7 Explanations and workload can drift

The global SHAP chart and local contributions describe different model variants. Label both clearly and treat SHAP as model behavior, not a causal effect. The cutoff averages 50 historical reviews, while the current quarter has 61 flags. Review distributions and workload each quarter; refresh scores, explanations and the PBIX from one version together. [S3, S4, S8]

Override policy and monitoring

Team policy design for a proposed pilot

Who may override and why

A designated reviewer who did not build the model may propose an override; the committee chair approves it. Add a below-cutoff bank using dated liquidity, funding or other documented evidence. Remove a flagged bank only when source verification, a known data error or case evidence supports removal. Disagreement with a score or concern about workload is not sufficient.

Keep the original model output intact

An override changes the review decision, not the stored score or model flag. Preserve both. Log RSSD ID, quarter, model version, score, original flag, direction, dated evidence, reason, proposer, approver and approval date. A removal must identify the driver being challenged. Reassess every override next quarter; a renewal needs current evidence.

Review each quarterly release

The pipeline owner reconciles coverage, checks missing inputs and tests label maturity. The reviewer compares flags, deposit coverage, concentration, form and size composition, score distributions and override reasons. Once outcomes mature, compute precision and recall with their eligible denominators. Recent unresolved outcomes are not successful predictions. Preserve data hashes, code commit and release date.

Escalate and suspend when needed

Suspend model-based recommendations for a label defect, a failed reconciliation or a model-dashboard version mismatch. Escalate workload beyond assumed capacity and repeated overrides involving one driver or bank type. The team retains escalation above 20 percent of the pre-override flagged list, counting unique banks added or removed; escalate any override when the original list is empty. This is a team-set policy design, not a validated or regulatory threshold.

Team decision and review scope

The team retains the 7 percent leverage, 3 percent noncurrent-loan and negative 0.5 percent annualized income/assets triggers, deduplicated 12 features, average 50-review budget and override policy. Loki completed selected source-value, label and dashboard spot checks, read this memo and accepted its conclusions. The validation log distinguishes these checks from automated tests and unreviewed items. This does not authorize live use or certify the full pipeline.

S1 Stress definition and time split
S2 Feature construction and corrected label logic
S3 Model fitting and feature selection
S4 Watchlist cutoff and export rules
S7 Federal Reserve May 2023 banking conditions and ratio definitions
S8 Automated v0.2 review with source hashes